Any Boolean function can be written in many equivalent algebraic forms. For example:

$$F = ABC + \overline{A}BC + AB\overline{C} \quad \text{simplifies to} \quad F = BC + AB$$

Simpler expression → fewer gates → faster/cheaper circuits.

Gate-level minimization: finds the optimal gate-level implementation of a Boolean function. Circuit complexity is directly tied to expression complexity.

"Simplest" is defined by two rules:

1. minimum number of product terms
2. minimum literals per term.

Karnaugh Map (K-Map): a visual rectangular grid of squares where each square represents one minterm. A 1 is placed wherever the function equals 1. The key property is that adjacent squares differ in exactly one variable. It is essentially a truth table rearranged for visual simplification.

We will look at 2 different ones today:

2-variable K-Map:

	$y = 0$	$y = 1$
$x = 0$	$m_0 = \overline{x}\overline{y}$	$m_1 = \overline{x}y$
$x = 1$	$m_2 = x\overline{y}$	$m_3 = xy$

3-variable K-Map:

	$yz{=}00$	$yz{=}01$	$yz{=}11$	$yz{=}10$
$x = 0$	m_0	m_1	m_3	m_2
$x = 1$	m_4	m_5	m_7	m_6

The 2-variable K-Map has 4 squares because $2^2 = 4$, and the 3-variable K-Map has 8 squares because $2^3 = 8$. The general formula for the number of squares needed for an N-variable K-Map is 2^N. Remember that minterms are in SOP form. When a minterm variable is set to 0, you complement that variable.

Each square represents the potential minterm rows in a truth table (based on the input combinations). To fill out a K-Map, place a 1 in every minterm square where the function output = 1. In other words, find the minterms and fill their associated box with a 1.

$F = xy$: only $m_3 = 1$. $G = x + y$: $m_1, m_2, m_3 = 1$ (at least one of x, y is 1).

$F = xy$

	$y = 0$	$y = 1$
$x = 0$	0	0
$x = 1$	0	**1**

$G = x + y$

	$y = 0$	$y = 1$
$x = 0$	0	**1**
$x = 1$	**1**	**1**

Switching over to the 3-variable K-Map now, notice the order of the input combinations above each column. The columns are not in binary order (00, 01, 10, 11). **Remember that adjacent squares differ**

in exactly one variable. If we look at the minterms for m1 and m3, x'y'z and x'yz share the same literals of x' and z, but the middle literal is the only one that is different (y' and y).

Core principle for minimizing: When you OR two adjacent minterm squares together, the differing variable cancels out, leaving a product term with one fewer literal. This provides us with the necessary foundation we need to effectively group minterms together. You may only group squares whose count is a power of 2: 1, 2, 4, or 8. Groups of 3, 5, 6, or 7 are not allowed.

Wrap-Around Adjacency: the left edge of a 3-varKM is adjacent to the right edge of a 3-varKM.

Steps to minimize using a K-Map:

1. Fill the map. Place 1s in every minterm square; leave others 0 or blank.
2. Find the largest groups. Group adjacent 1s in power-of-2 rectangles (1, 2, 4, 8). Bigger is better. Wrap-around counts.
3. Cover all 1s when possible. A 1 can belong to multiple groups.
4. No redundant groups. Don't add a group if all its 1s are already covered.
5. Read each group. Keep variables that are constant across the group (unprimed if always 1; primed if always 0). Drop variables that change.
6. OR the terms. The final answer is the sum of all group terms.

Ex)

Simplify: $F(x, y, z) = \sum(2, 3, 4, 5)$

Step 1: Fill the map. Place 1s in squares m_2, m_3, m_4, m_5.

	$yz=00$	$yz=01$	$yz=11$	$yz=10$
$x=0$	0	0	**1**	**1**
$x=1$	**1**	**1**	0	0

Step 2: Identify groups.

- **Blue group**: m_2 and m_3 (top-right two squares)
- **Green group**: m_4 and m_5 (bottom-left two squares)

Step 3: Read each group.

Blue group (m_2, m_3)*:* $m_2 = \overline{x}y\overline{z}$ and $m_3 = \overline{x}yz$.

- $x = 0$ throughout $\Rightarrow$ keep $\overline{x}$
- $y = 1$ throughout $\Rightarrow$ keep y
- z changes (0 then 1) $\Rightarrow$ **drop** z
- Result: $\overline{x}y$

Green group (m_4, m_5)*:* $m_4 = x\overline{y}\overline{z}$ and $m_5 = x\overline{y}z$.

- $x = 1$ throughout $\Rightarrow$ keep x
- $y = 0$ throughout $\Rightarrow$ keep $\overline{y}$
- z changes $\Rightarrow$ **drop** z
- Result: $x\overline{y}$

Final answer:

$$\boxed{F = \overline{x}y + x\overline{y}}$$

Simplify: $F(x, y, z) = \sum(3, 4, 6, 7)$
Step 1: Fill the map.

	$yz{=}00$	$yz{=}01$	$yz{=}11$	$yz{=}10$
$x = 0$	0	0	**1**	0
$x = 1$	**1**	0	**1**	**1**

Step 2: Identify groups.

- **Blue group**: m_3 and m_7 — these are in the same column ($yz = 11$), vertically adjacent.
  $m_3 = \overline{x}yz$, $m_7 = xyz$. x changes; $y = 1$, $z = 1$ throughout $\Rightarrow$ term: yz.
- **Green group**: m_4 and m_6 — these are in the left edge column ($yz = 00$) and right edge column ($yz = 10$). They **wrap around**!
  $m_4 = x\overline{y}\overline{z}$, $m_6 = xy\overline{z}$. y changes; $x = 1$, $z = 0$ throughout $\Rightarrow$ term: $x\overline{z}$.

Final answer:

$$\boxed{F = yz + x\overline{z}}$$

Ex)

Simplify: $F(x, y, z) = \sum(0, 2, 4, 5, 6)$
Step 1: Fill the map.

	$yz{=}00$	$yz{=}01$	$yz{=}11$	$yz{=}10$
$x = 0$	**1**	0	0	**1**
$x = 1$	**1**	**1**	0	**1**

Step 2: Identify groups.

Blue group of 4: m_0, m_2, m_4, m_6 (all four corners, two direct and two via wrap-around). x and y both change $\Rightarrow$ drop both. $z = 0$ throughout $\Rightarrow$ keep $\overline{z}$. **Term:** $\overline{z}$
Green group of 2: *m_4 and m_5* (m_4 is reused from the blue group — that's OK!). $x = 1$ and $y = 0$ throughout; z changes $\Rightarrow$ drop z. **Term:** $x\overline{y}$

$$\boxed{F = \overline{z} + x\overline{y}}$$

Ex)